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Contents

Intellectual Property Rights 4

Foreword 4

1 Scope 5

2 References 5

3 Definitions and abbreviations 7
3.1 Definitions 7
3.2 Abbreviations 7

4 enhanced Multi-Level Precedence and Pre-emption (eMLPP) 7
4.1 Normal operation 7
4.1.1 Mobile originated calls 7
4.1.2 Mobile terminated calls 9
4.1.3 Called party pre-emption for mobile terminated calls 9
4.1.4 Group call or broadcast call, calling mobile station 9
4.1.5 Group or broadcast call, called mobile stations 11
4.2 Registration 11
4.3 Erasure 11
4.4 Activation, deactivation 12
4.5 Interrogation 12

Annex A (informative): Change Request History 13

History 14

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Foreword

This European Standard (Telecommunications series) has been produced by the Special Mobile Group (SMG).

The present document specifies the stage three description of the enhanced Multi-Level Precedence and Pre-emption Service (eMLPP) within the digital cellular telecommunications system.

The contents of the present document is subject to continuing work within SMG and may change following formal SMG approval. Should SMG modify the contents of the present document it will be re-released with an identifying change of release date and an increase in version number as follows:

Version 6.x.y

where:

6 indicates Release 1997 of GSM Phase 2+

x the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

y the third digit is incremented when editorial only changes have been incorporated in the specification.

Proposed national transposition dates	
Date of adoption of this EN:	3 September 1999
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Date of latest publication of new National Standard or endorsement of this EN (dop/e):	30 June 2000
Date of withdrawal of any conflicting National Standard (dow):	30 June 2000

1 Scope

The present document specifies the procedures used at the radio interface (Reference point Um as defined in GSM 04.02) for normal operation, invocation, registration and interrogation of the enhanced Multi-Level Precedence and Pre-emption Service (eMLPP) supplementary service. Provision and withdrawal of supplementary services is an administrative matter between the mobile subscriber and the service provider and cause no signalling on the radio interface.

In GSM 04.10 the general aspects of the specification of supplementary services at the layer 3 radio interface are given.

GSM 04.80 specifies the formats and coding for the supplementary services.

Definitions and descriptions of supplementary services are given in GSM 02.04 and GSM 02.8x and 02.9x-series. GSM 02.67 is related specially to eMLPP.

Technical realization of supplementary services is described in GSM 03.11 and GSM 03.8x and 03.9x-series.

GSM 03.67 is related specially to eMLPP.

The procedures for Call Control, Mobility Management and Radio Resource management at the layer 3 radio interface are defined in GSM 04.07 and GSM 04.08.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.
- For this Release 1997 document, references to GSM documents are for Release 1997 versions (version 6.x.y).

[1] GSM 01.04: "Digital cellular telecommunications system (Phase 2+); Abbreviations and acronyms".

[2] GSM 02.04: "Digital cellular telecommunications system (Phase 2+); General on supplementary services".

[3] GSM 02.67: "Digital cellular telecommunications system (Phase 2+); enhanced Multi-Level Precedence and Pre-emption service (eMLPP) - Stage 1".

[4] GSM 02.81: "Digital cellular telecommunications system (Phase 2+); Line identification supplementary services - Stage 1".

[5] GSM 02.82: "Digital cellular telecommunications system (Phase 2+); Call Forwarding (CF) supplementary services - Stage 1".

[6] GSM 02.83: "Digital cellular telecommunications system (Phase 2+); Call Waiting (CW) and Call Hold (HOLD) supplementary services - Stage 1".

[7] GSM 02.84: "Digital cellular telecommunications system (Phase 2+); MultiParty (MPTY) supplementary services - Stage 1".

[8] GSM 02.85: "Digital cellular telecommunications system (Phase 2+); Closed User Group (CUG) supplementary services - Stage 1".

[9] GSM 02.86: "Digital cellular telecommunications system (Phase 2+); Advice of Charge (AoC) Supplementary Services - Stage 1".

[10] GSM 02.88: "Digital cellular telecommunications system (Phase 2+); Call Barring (CB) supplementary services - Stage 1".

[11] GSM 02.90: "Digital cellular telecommunications system (Phase 2+); Stage 1 description of Unstructured Supplementary Service Data (USSD)".

[12] GSM 02.91: "Digital cellular telecommunications system (Phase 2+); Explicit Call Transfer (ECT)".

[13] GSM 03.11: "Digital cellular telecommunications system (Phase 2+); Technical realization of supplementary services".

[14] GSM 03.67: "Digital cellular telecommunications system (Phase 2+); enhanced Multi-Level Precedence and Pre-emption service (eMLPP) - Stage 2".

[15] GSM 03.68: "Digital cellular telecommunications system (Phase 2+);Voice Group Call Service (VGCS) - Stage 2".

[16] GSM 03.69: "Digital cellular telecommunications system (Phase 2+);Voice Broadcast Service (VBS) - Stage 2".

[17] GSM 03.81: "Digital cellular telecommunications system (Phase 2+); Line identification supplementary services - Stage 2".

[18] GSM 03.82: "Digital cellular telecommunications system (Phase 2+); Call Forwarding (CF) supplementary services - Stage 2".

[19] GSM 03.83: "Digital cellular telecommunications system (Phase 2+); Call Waiting (CW) and Call Hold (HOLD) supplementary services - Stage 2".

[20] GSM 03.84: "Digital cellular telecommunications system (Phase 2+); MultiParty (MPTY) supplementary services - Stage 2".

[21] GSM 03.85: "Digital cellular telecommunications system (Phase 2+); Closed User Group (CUG) supplementary services - Stage 2".

[22] GSM 03.86: "Digital cellular telecommunications system (Phase 2+); Advice of Charge (AoC) supplementary services - Stage 2".

[23] GSM 03.88: "Digital cellular telecommunications system (Phase 2+); Call Barring (CB) supplementary services - Stage 2".

[24] GSM 03.90: "Digital cellular telecommunications system (Phase 2+); Unstructured supplementary services operation - Stage 2".

[25] GSM 03.91: "Digital cellular telecommunications system (Phase 2+); Explicit Call Transfer (ECT) supplementary service - Stage 2".

[26] GSM 04.02: "Digital cellular telecommunications system (Phase 2+); GSM Public Land Mobile Network (PLMN) access reference configuration".

[27] GSM 04.07: "Digital cellular telecommunications system (Phase 2+); Mobile radio interface signalling layer 3; General aspects".

[28] GSM 04.08: "Digital cellular telecommunications system (Phase 2+); Mobile radio interface layer 3 specification".

[29] GSM 04.10: "Digital cellular telecommunications system (Phase 2+); Mobile radio interface layer 3 Supplementary services specification; General aspects".

[30] GSM 04.80: "Digital cellular telecommunications system (Phase 2+); Mobile radio interface layer 3 supplementary services specification Formats and coding".

3 Definitions and abbreviations

3.1 Definitions

Definitions used in the present document are defined in GSM 02.67 and GSM 03.67.

3.2 Abbreviations

Abbreviations used in the present document are listed in GSM 01.04.

For the purposes of the present document, the following abbreviations apply:

eMLPP	enhanced MLPP
MLPP	Multi-Level Precedence and Pre-emption
VBS	Voice Broadcast Service
VGCS	Voice Group Call Service

4 enhanced Multi-Level Precedence and Pre-emption (eMLPP)

4.1 Normal operation

NOTE: In the call related messages only the additional information for the eMLPP supplementary service is shown in the figures.

4.1.1 Mobile originated calls

The mobile station can indicate the priority of each call initiated. If no priority is indicated by the user or a non-compatible mobile station is used then the default priority level shall be applied which is stored in the VLR. The selection of priority shall be an MMI function.

For mobile originated calls, the corresponding message flows are shown in figure 1.

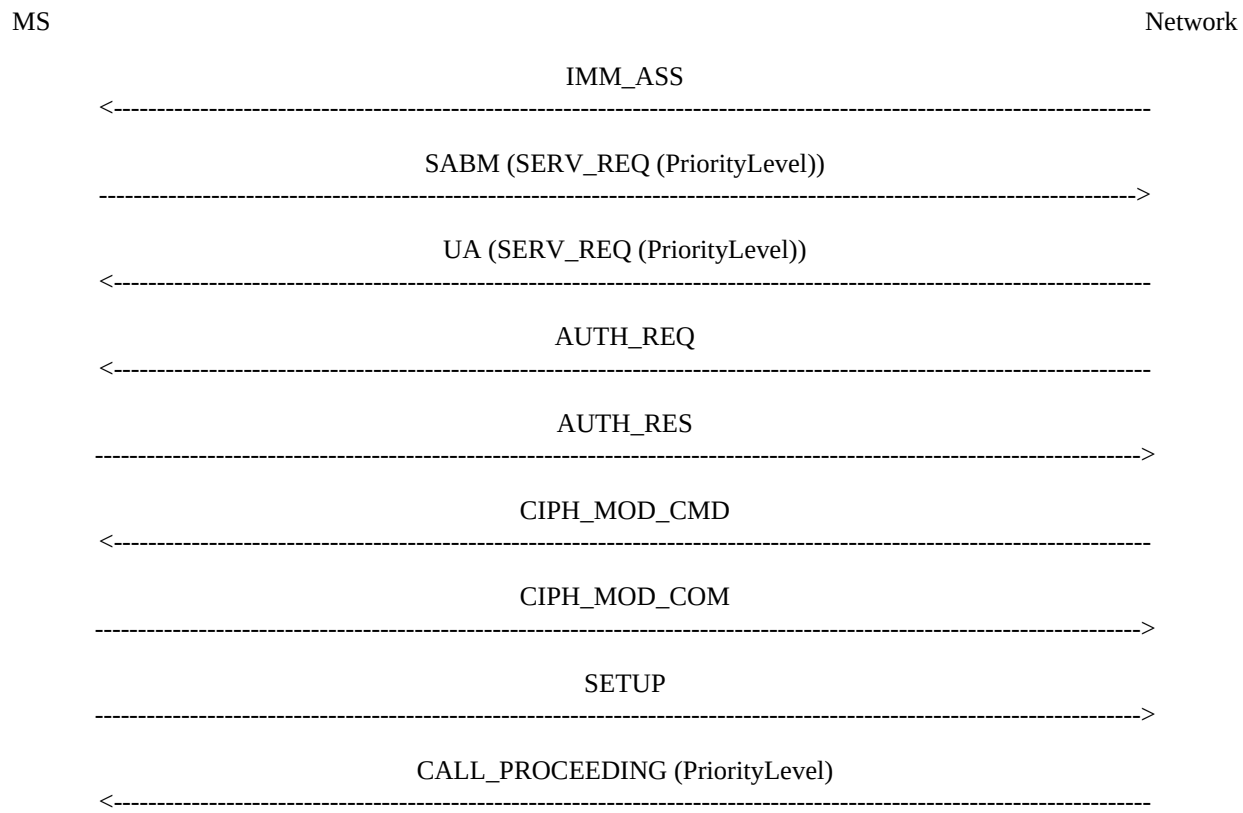

Figure 1: Signalling information required for the prioritisation at mobile originating call establishment

IMM_ASS: Standard message which is sent if no RR connection was already established.

SABM (SERV_REQ (PriorityLevel)): L3-MM CM SERVICE REQUEST where the priority level information element is provided in addition if a priority selection is performed by the user. In case of no priority selection or use of a non-compatible mobile station the mobile station shall send a service request message without priority level information element and the network shall apply a default priority to the request. The message may be piggybacked in a SABM if no RR connection was already established.

UA (SERV_REQ (PriorityLevel): Standard message to acknowledge the layer 2 link which is sent if no RR connection was already established. The priority level is the same as received by the network.

AUTH_REQ: Standard message which is sent if the network applies authentication as shown in figure 1. If not, the network will sent a standard CM_SERVICE_ACCEPT message.

AUTH_RES: Standard message which is sent if the network applies authentication.

CIPH_MOD_CMD: Standard message which is sent if the network applies ciphering as shown in figure 1.

CIPH_MOD_COM: Standard message which is sent if the network applies ciphering.

SETUP: Standard message.

CALL_PROCEEDING: The network shall include the assigned priority level in a CALL_PROCEEDING message if no priority level was indicated in the CM_SERVICE_REQUEST message or the assigned priority level is different to the one requested.

4.1.2 Mobile terminated calls

For a mobile terminated call the priority level is indicated by the calling subscriber if MLPP is used. The mobile station shall be paged in the normal manner, but with the paging messages also containing the priority level of the call. In addition, the priority level shall be included in the set-up message.

NOTE: Mobile stations in idle mode which are going to respond to a paging message do not need to analyse the priority level in the paging request message but can take the priority level provided in the set-up message. Mobile stations in group receive mode [or with active GPRS modes or other future services ongoing] need to analyse the priority level in the paging request message in order to decide to respond to the paging request.

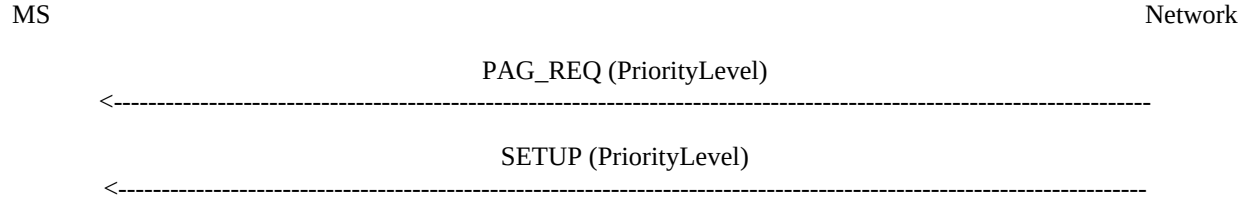

Figure 2: Signalling information required for the prioritisation at mobile terminating call establishment without called-party pre-emption.

PAG_REQ: Paging message including the related priority level to be applied.

SETUP: Modified SETUP message with an indication of the priority level.

4.1.3 Called party pre-emption for mobile terminated calls

In the case where the called subscriber has a subscription for eMLPP and for CW, the mobile station shall be informed of the priority of the new call together with the CW indication. On reception of the set-up message the compatible mobile station decides on called party pre-emption. If called party pre-emption applies, the mobile station shall automatically accept the waiting call and send a hold message to the network. If a hold acknowledge is received, the waiting call is accepted. If a hold reject is received for any reason, e.g. there is no subscription for hold, the other call shall be released and the waiting call accepted. If the ongoing call is not a TS11 call, the mobile station should not send a hold message to the network but release the call and accept the waiting call.

If no pre-emption applies or the called party is using a non compatible mobile station, CW will be used as normal.

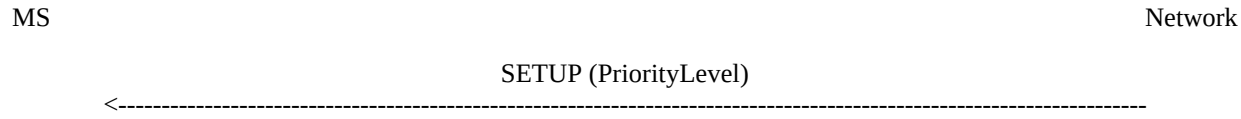

Figure 3: Signalling information required for the called-party pre-emption in case of use of a compatible mobile station

SETUP: Modified SETUP message with an indication of the priority level.

4.1.4 Group call or broadcast call, calling mobile station

Within each set of voice group call or voice broadcast call attributes stored in the GCR as defined in GSM 03.68 and GSM 03.69, respectively, a priority level is included if eMLPP is applied. The priority level will be provided by the GCR to the MSC together with the call attributes.

For VGCS or VBS establishment, the calling mobile station may indicate a priority level in the service request as in subclause 4.1.1. This priority level can be applied for the dedicated link of the calling mobile station as long as no different priority level in provided by the GCR. If this happens, the priority level provided by the GCR shall also be applied to the dedicated link of the calling mobile station.

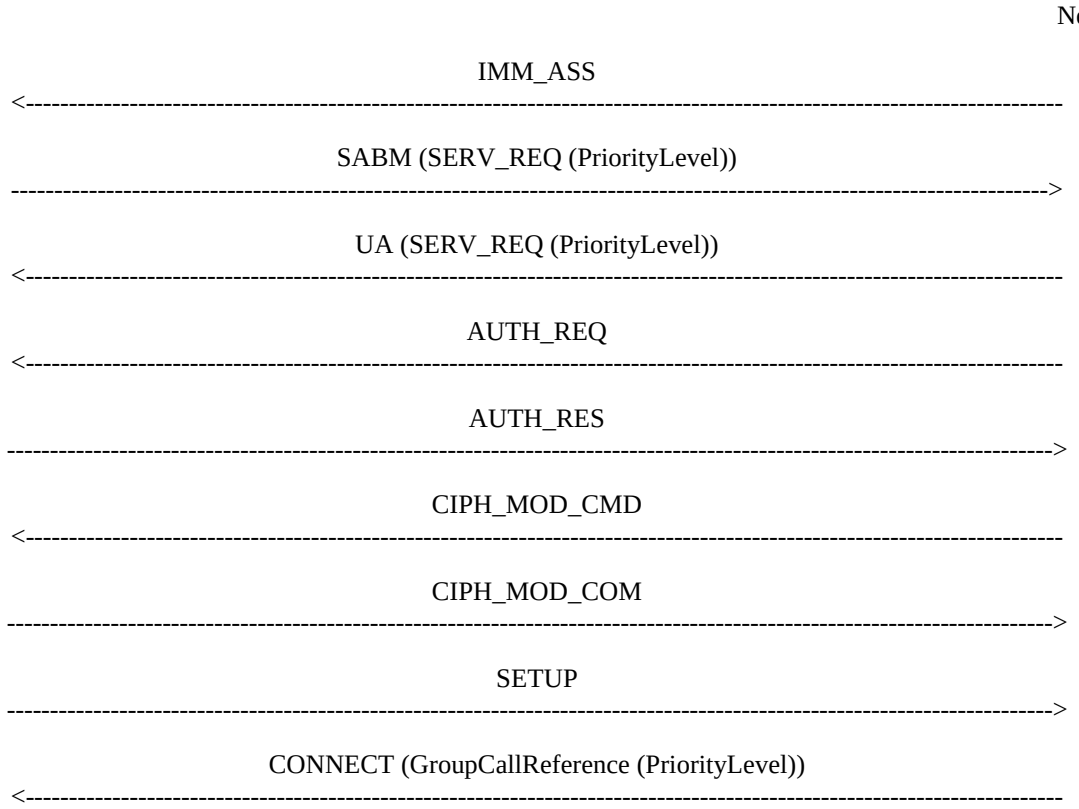

Figure 4: Signalling information between the network and the calling mobile station required for the prioritisation in case of a VGCS or VBS call

IMM_ASS: Standard message which is sent if no RR connection was already established.

SABM (SERV_REQ (PriorityLevel)): L3-MM CM SERVICE REQUEST where the priority level information element is provided in addition if a priority selection is performed by the user. In case of no priority selection or use of a non-compatible mobile station the mobile station shall send a service request message without priority level information element and the network shall apply a default priority to the request. The message may be piggybacked in a SABM if no RR connection was already established.

UA (SERV_REQ (PriorityLevel)): Standard message to acknowledge the layer 2 link which is sent if no RR connection was already established. The priority level is the same as received by the network.

AUTH_REQ: Standard message which is sent if the network applies authentication as shown in figure 4. If not, the network will sent a standard CM_SERVICE_ACCEPT message.

AUTH_RES: Standard message which is sent if the network applies authentication.

CIPH_MOD_CMD: Standard message which is sent if the network applies ciphering as shown in figure 4.

CIPH_MOD_COM: Standard message which is sent if the network applies ciphering.

SETUP: Standard message.

CONNECT: Information to the mobile station that the VGCS or VBS call is established with the related group or broadcast call reference as the connected number. The group or broadcast call reference includes the priority level applied for the group or broadcast call in the network. This priority level can be different to the one indicated in the CM_SERVICE_REQUEST.

4.1.5 Group or broadcast call, called mobile stations

Within each set of voice group call or voice broadcast call attributes stored in the GCR as defined in GSM 03.68 and GSM 03.69, respectively, a priority level is included if eMLPP is applied. The priority level will be provided by the GCR to the MSC together with the call attributes.

The priority level shall be indicated together with the related notification messages and treated in the mobile station as defined in GSM 03.68 and GSM 03.69, respectively.

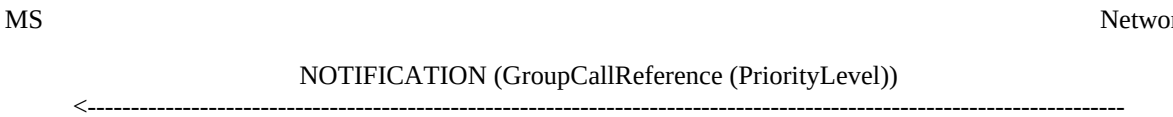

Figure 5: Signalling information between the network and the called mobile stations required for the prioritisation in case of a VGCS or VBS call

NOTIFICATION: L3 RR message NOTIFICATION/NCH_TYPE1 or NOTIFICATION/NCH_TYPE2 or NOTIFICATION/FACCH or NOTIFICATION/SACCH containing a group call reference or broadcast call reference of a notified voice group or broadcast call which includes a related priority level for that call.

4.2 Registration

The default eMLPP priority level has to be registered in the network:

An eMLPP registration request from a mobile user shall include the SS-Code of the eMLPP service and the default priority level.

If the registration is successful, the default eMLPP priority level will be registered. The network will then send a return result indicating acceptance of the request, including the default eMLPP priority level.

If the system cannot accept a registration request, a corresponding error indication is returned to the served mobile subscriber that eMLPP registration was not successful. Error values are specified in GSM 04.80.

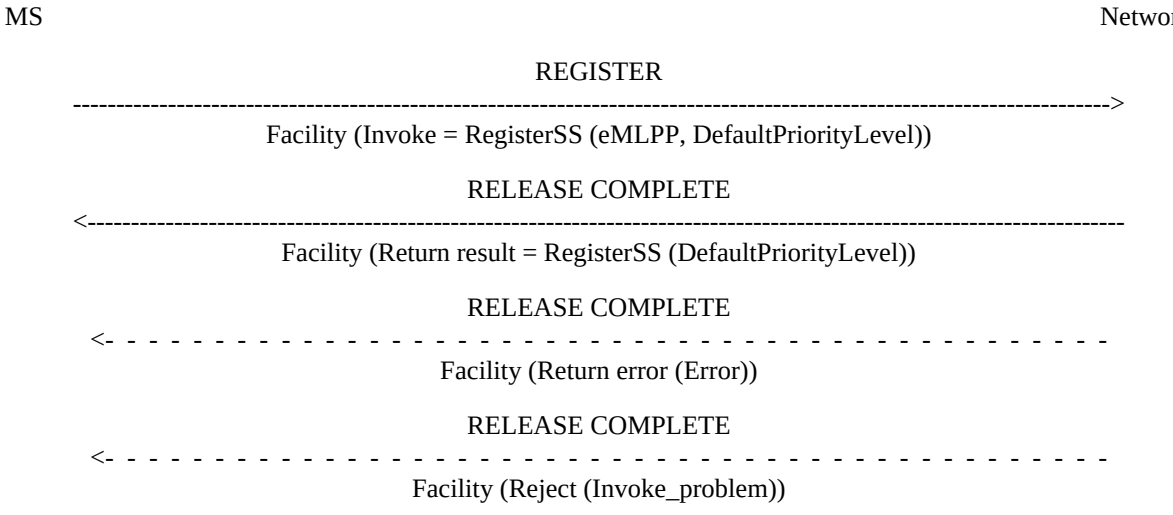

Figure 6: Registration default priority level

4.3 Erasure

A previous registration can be erased in one of two ways:

- the subscriber can register information for eMLPP for a new default priority level, thus causing the previous registration of eMLPP to be overridden;
- all information is erased as a result of withdrawal of the supplementary service (administrative handling).

4.4 Activation, deactivation

Activation and deactivation of the supplementary service enhanced Multi-Level Precedence and Pre-emption are not applicable.

4.5 Interrogation

The interrogation procedure enables the mobile subscriber to obtain information about data stored in the PLMN. The eMLPP service subscriber may interrogate the maximum priority level he can use and the actual default priority level.

If the service is provisioned, the network shall sent a return result including the SS-Status parameter and the maximum priority level which the service subscriber is allowed to use and the actual default priority level.

If the service is not provisioned, the network shall send a return result including the SS-Status parameter.

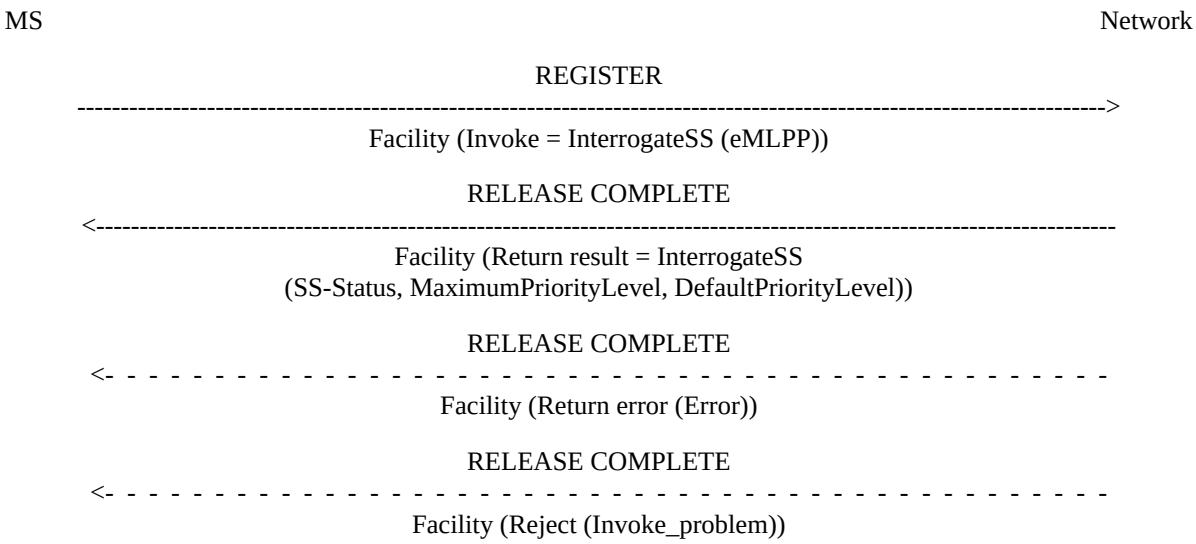

Figure 7: Interrogation of the maximum and default priority levels

Annex A (informative): Change Request History

Status of Technical Specification GSM 04.67		
Date	**Version**	**Remarks**
		No phase 1 version
December 1994	version 0.0.0	The first draft
October 1996	version 0.3.3	outcome SMG3-WPB; to SMG3 for information
October 1996	version 1.0.0	to SMG#20 for information
October 1996	version 5.0.0	TS approved by SMG#20
December 1996	version 5.0.0	GTS converted to draft prETS 300 947
May 1997	version 5.0.1	ETS 300 947 first edition
January 1999	version 6.0.0	Release 1997 version
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